WORKED SOLUTION

STEP I 2001, Question 13

Probability and Statistics • Conditional Poisson Distribution

STEP I 2001, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q13

The key idea

Conditioning on the total number of breakages turns a Poisson question into a binomial one. If the four students break plates independently at the same rate, then once we know how many plates were broken altogether, each broken plate is equally likely to have been broken by any of the four, independently of the others. So the mathematician's share of the broken plates is binomial, and the unknown λ disappears.

Solution

Setting up

Let M be the number of plates broken by the mathematician, and let R be the total number broken by the other three students. We are told that each student's breakages are $\mathrm{Po}(\lambda)$, independently of the others. So $M \sim \mathrm{Po}(\lambda)$, and R, being a sum of three independent $\mathrm{Po}(\lambda)$ variables, is $\mathrm{Po}(3\lambda)$. Also M and R are independent, because R depends only on the other three students. The total is $T = M + R \sim \mathrm{Po}(4\lambda)$.

? **Ask yourself** Why is a sum of independent Poisson variables Poisson again?

For independent $X \sim \mathrm{Po}(\alpha)$ and $Y \sim \mathrm{Po}(\beta)$, sum over the ways of splitting n:

$$\mathrm{P}(X+Y=n) = \sum_{k=0}^{n} \frac{\mathrm{e}^{-\alpha}\alpha^k}{k!} \cdot \frac{\mathrm{e}^{-\beta}\beta^{n-k}}{(n-k)!} = \frac{\mathrm{e}^{-(\alpha+\beta)}}{n!} \sum_{k=0}^{n} \binom{n}{k} \alpha^k \beta^{n-k} = \frac{\mathrm{e}^{-(\alpha+\beta)}(\alpha+\beta)^n}{n!},$$

by the binomial theorem. So $X + Y \sim \mathrm{Po}(\alpha+\beta)$. Applying this twice gives $R \sim \mathrm{Po}(3\lambda)$, and once more gives $T \sim \mathrm{Po}(4\lambda)$.

The conditional distribution of M

For $m = 0, 1, \ldots, 5$, the event $\{M = m,\ T = 5\}$ is the same as $\{M = m,\ R = 5 - m\}$. By independence of M and R,

$$\begin{aligned}
\mathrm{P}(M = m \mid T = 5) &= \frac{\mathrm{P}(M = m)\,\mathrm{P}(R = 5 - m)}{\mathrm{P}(T = 5)} \\
&= \frac{\dfrac{\mathrm{e}^{-\lambda}\lambda^m}{m!} \cdot \dfrac{\mathrm{e}^{-3\lambda}(3\lambda)^{5-m}}{(5-m)!}}{\dfrac{\mathrm{e}^{-4\lambda}(4\lambda)^5}{5!}} \\
&= \frac{5!}{m!\,(5-m)!} \cdot \frac{3^{5-m}}{4^5} = \binom{5}{m} \left(\tfrac{1}{4}\right)^m \left(\tfrac{3}{4}\right)^{5-m}.
\end{aligned}$$

The exponentials cancel, and so does every power of λ (there are λ^5 on top and bottom). So, given $T = 5$, we have $M \sim \mathrm{B}(5, \frac{1}{4})$, whatever the value of λ.

The answer

Summing the binomial probabilities for $m = 3, 4, 5$:

$$\begin{aligned}
\mathrm{P}(M \geq 3 \mid T = 5) &= \binom{5}{3}\frac{3^2}{4^5} + \binom{5}{4}\frac{3}{4^5} + \binom{5}{5}\frac{1}{4^5} \\
&= \frac{10 \times 9 + 5 \times 3 + 1}{1024} = \frac{106}{1024} = \frac{53}{512}.
\end{aligned}$$

So the probability that the mathematician broke three or more of the five plates is $\dfrac{53}{512} \approx 0.104$.

Checks

- The six conditional probabilities $\binom{5}{m}3^{5-m}/4^5$ add up to $(1+3)^5/4^5 = 1$, as they must.
- Symmetry: by the same argument each student's expected share of the five plates is $5 \times \frac{1}{4} = \frac{5}{4}$, and the four shares add up to 5.
- We computed $\mathrm{P}(M \geq 3 \mid T = 5)$ directly from the four Poisson distributions (summing over every way of sharing five breakages among four students) for $\lambda = 0.3, 1, 2.7$ and 5, and got $0.103515625 = 53/512$ each time. A simulation of 4×10^5 rounds of washing up gave about 0.105 among the rounds with exactly five breakages.

Common slips

- Leaving λ in the answer. It must cancel; if it doesn't, the conditioning has gone wrong.

- Dividing by $\mathrm{P}(M+R=5)$ but treating R as $\mathrm{Po}(\lambda)$ instead of $\mathrm{Po}(3\lambda)$, which would amount to only two students.
- Forgetting the binomial coefficients, or treating the event as "exactly three" rather than "three or more".

Beyond STEP

This is a small instance of a general fact: if $X_1, \ldots, X_k$ are independent Poisson variables with means $\lambda_1, \ldots, \lambda_k$, then given their sum is n the vector $(X_1, \ldots, X_k)$ is multinomial with probabilities $\lambda_i / \sum \lambda_j$. Run backwards, it says that if each of a $\mathrm{Po}(\mu)$ number of events is independently labelled with one of k types, the counts of each type are independent Poisson variables. This "Poisson splitting" is a basic tool in the study of Poisson processes. Blitzstein and Hwang, *Introduction to Probability*, discusses both directions with many examples.

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